Description

This project focuses on analyzing the inbound call operations of **ABC Insurance Company** to understand customer calling behavior and service performance. The dataset spans **23 days of call center activity** and captures detailed information about customer interactions with the inbound support team.

The primary objective of this analysis is to study **call volume trends across different time buckets**, calculate **average call duration**, and design an **effective manpower plan**. By applying data-driven techniques, the project aims to reduce the existing **call abandonment rate of approximately 30%** and improve overall **customer experience (CX)**.

Key Objectives

- ❑ Analyze inbound call trends across hourly time buckets
- ❑ Understand agent workload using average call duration
- ❑ Identify peak and low call volume periods
- ❑ Propose an optimized manpower allocation strategy
- ❑ Reduce call abandonment to **10%** while maintaining service quality

Business Understanding

Inbound call centers act as a **critical customer touchpoint**, especially in the insurance sector where customers require timely assistance. High call abandonment leads to **customer dissatisfaction, loss of trust, and negative brand perception**. Efficient manpower planning ensures that customer calls are handled promptly, reducing wait time and improving service quality while controlling operational costs.

Dataset Overview

The dataset used for this analysis contains detailed inbound call information, including:
Agent Name and Agent ID
Queue Time (in seconds)
Call Date and Time
Call Duration (in seconds)
Call Status (Answered, Abandoned, Transferred)
Hour-wise Time Buckets
Data Period: 23 days of inbound call activity

This data enables a comprehensive analysis of call patterns and agent performance, forming the basis for effective manpower planning

Approach

A structured and data-driven approach was followed to analyze the inbound call data and propose an optimized manpower plan:

- **Data Understanding & Cleaning**
  - Reviewed the dataset to understand all variables such as call duration, queue time, call status, and time of call.
  - Cleaned the data by checking for missing values, incorrect formats, and inconsistencies.
- **Time Bucket Creation**
  - Converted call timestamps into **hour-wise time buckets** (e.g., 9–10, 10–11) to analyze call patterns throughout the day.
- **Average Call Duration Analysis**
  - Calculated the **average call duration** for each time bucket using pivot tables.
  - Used this metric to estimate agent handling capacity.
- **Call Volume Trend Analysis**
  - Analyzed total inbound calls for each time bucket.
  - Visualized call volume trends using charts to identify peak and off-peak hours.

- **Day Shift Manpower Planning (9 AM – 9 PM)**
    - Calculated the **minimum number of agents required per time bucket** to ensure at least **90% of calls are answered**, limiting the abandon rate to 10%.
- **Night Shift Manpower Estimation (9 PM – 9 AM)**
    - Estimated night call volume as **30% of day calls**, as per the given assumption.
    - Distributed night calls across hourly time buckets and calculated required agents.
- **Real-World Adjustments**
    - Incorporated practical assumptions such as agent working days, unplanned leaves, break time, and productive call time to calculate actual monthly headcount requirements.

Tech stack

❖ **Microsoft Excel (2022)**

- Used for data cleaning and preparation
- Created pivot tables to analyze:
    - Call volume trends
    - Average call duration by time bucket
- Applied formulas for:
    - Agent capacity calculation
    - Manpower planning and abandonment control
- Used charts for visual representation of call volume patterns

❖ **Microsoft PowerPoint**

- Used to present analysis, insights, and recommendations
- Enabled clear communication of business findings to stakeholders

❖ **Google Drive**

Used to store and share:

- Excel analysis file
- Final project report (PDF)

- Enabled hyperlinking of files in the presentation

Here is the analysis file that was done in the excel file as per mentioned in the details of the project

report

key Insights from the Analysis

- **Call volume is not uniform throughout the day**. Peak call traffic is observed during late morning and evening hours, indicating the need for higher agent availability during these periods.
- **Average call duration remains relatively consistent** across time buckets (approximately 3.3 minutes), allowing reliable estimation of agent handling capacity.
- The current **abandonment rate is high (~30%)**, primarily due to insufficient agent availability during peak hours.

•**Critical Observation (Data Quality & Business Insight):**
For calls recorded **until 13th January**, several records show **Agent Name / Agent ID as NULL**.
These calls were **automatically marked as abandoned**, indicating that **no agents were available at that time** to handle incoming calls.

•This issue is **especially prominent during night hours**, where the absence of agents resulted in **100% call abandonment**, leading to poor customer services

•Proper manpower planning, both during **day shifts and night shifts**, has a significant impact on reducing abandonment and improving service levels.

Result

- Successfully analyzed **23 days of inbound call data** to understand call volume trends and customer calling behavior across different time buckets.
- Calculated the **average call duration** for each time bucket, which helped determine agent handling capacity and supported accurate manpower planning.
- Identified **peak and off-peak hours**, enabling a data-driven approach to agent allocation rather than fixed staffing.
- Designed an **optimized manpower plan for day shift (9 AM–9 PM)** that reduces the call abandonment rate from **approximately 30% to 10%** by ensuring that at least **90% of calls are answered**.

- ❑ Proposed a **night shift manpower model (9 PM–9 AM)** based on realistic assumptions, addressing the issue of unanswered calls and improving **24×7 scustomer support availability**.
- ❑ Incorporated **real-world operational constraints** such as agent working days, unplanned leaves, break time, and productive call time, improving the practicality and accuracy of staffing recommendations.
- ❑ Gained a strong understanding of how **call volume trends, average handling time, and workforce shrinkage** directly impact customer experience and operational efficiency in a call center environment